



ASCHAM SCHOOL

2025 YEAR 12

MATHEMATICS EXTENSION 2

TRIAL EXAM

GENERAL INSTRUCTIONS

Reading time – 10 minutes
Working time – 180 minutes
Use black pen, non-erasable
NESA-approved calculators may be used
Reference Sheet is provided

Total Marks - 100

Section A – Multiple Choice (1 mark each)

Attempt Questions 1 to 10.

Select answers on the separate multiple-choice sheet provided.

Write your NESA number on the multiple-choice sheet.

Section B – Questions 11 – 16 (15 marks each)

Start each question in a new booklet.

If you use a second booklet for a question, place it inside the first.

Label extra booklets for the same question as, for example, Q11-2 etc.

Write your NESA number and question number on each booklet.

Section A - Multiple choice (10 marks)

(Mark the correct answer on the sheet provided.)

1. Given $0 < \theta < \frac{\pi}{2}$, what is the magnitude of the vector

$$(\cos \theta)\underline{i} + (\sin \theta)\underline{j} + (\tan \theta)\underline{k} \text{ ?}$$

A) 1

B) $\operatorname{cosec} \theta$

C) $\cot \theta$

D) $\sec \theta$

2. Which is the **negation** of the following statement?

“If snow is falling, then I will not be at the park.”

A) If snow is falling, then I will be at the park.

B) If snow is not falling, then I will be at the park.

C) Snow is falling and I will be at the park.

D) Snow is not falling and I will be at the park.

3. A particle whose displacement is x moves in simple harmonic motion such that $\ddot{x} = -4x$.

What is x as a function of t if initially $x = 3$ and $\dot{x} = -6\sqrt{3}$?

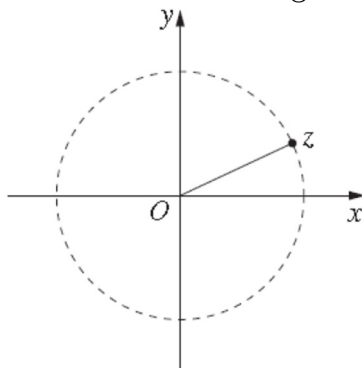
A) $x = 3 \cos 2t - 3\sqrt{3} \sin 2t$

B) $x = 3 \sin 2t - 3\sqrt{3} \cos 2t$

C) $x = 3 \cos 2t + 3\sqrt{3} \sin 2t$

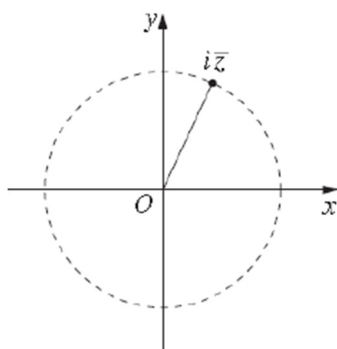
D) $x = 3 \sin 2t + 3\sqrt{3} \cos 2t$

4. The complex number z is shown on the Argand diagram below.

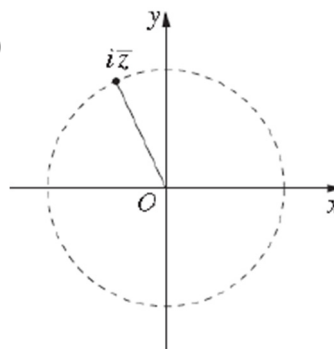


Which of the following best represents $i\bar{z}$?

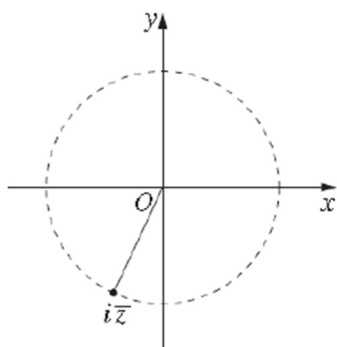
A)



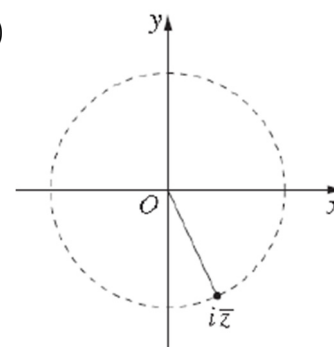
B)



B)



D)



5. P , Q and R are three collinear points with position vectors $\underline{p}$, $\underline{q}$ and $\underline{r}$ respectively. Q lies between P and R such that $PQ : QR = 2 : 1$.

The position vector of R is given by:

A) $\underline{r} = \frac{1}{2}\underline{q} + \frac{3}{2}\underline{p}$

B) $\underline{r} = \frac{3}{2}\underline{p} - \frac{1}{2}\underline{q}$

C) $\underline{r} = \frac{3}{2}\underline{q} - \frac{1}{2}\underline{p}$

D) $\underline{r} = \frac{1}{2}\underline{p} + \frac{3}{2}\underline{q}$

6. For which of these statements, for positive integers x and y , is the **converse** necessarily true?

A) If x and y are odd, then $x + y$ is even.

B) If x and y are odd, then $x^2 + y^2$ is even.

C) If x and y are odd, then xy is odd.

D) If x and y are odd, then x^y is odd.

7. Which of the following lines passes through $(0, 1, -1)$?

A) $\vec{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

B) $\vec{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

C) $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

D) $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

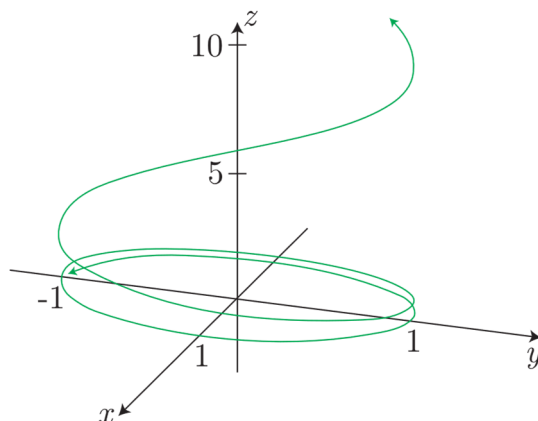
8. Which of these inequalities is **false**?

A) $\int_1^2 \frac{1}{1+x} dx < \int_1^2 \frac{1}{x} dx$

B) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin x}{x} dx < \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{x} dx$

C) $\int_1^2 e^{-x^2} dx < \int_0^1 e^{-x^2} dx$

D) $\int_0^{\frac{\pi}{4}} \tan^2 x dx < \int_0^{\frac{\pi}{4}} \tan^3 x dx$



- A) $\underline{r}(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (t)\underline{k}$
- B) $\underline{r}(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (e^{-0.5t})\underline{k}$
- C) $\underline{r}(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (\ln t)\underline{k}$
- D) $\underline{r}(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (\cos 2t)\underline{k}$

10. Let $w = e^{i\frac{2\pi}{5}}$.
What is the value of $(1-w)(1-w^2)(1-w^3)(1-w^4)$?

- A) -5
- B) -4
- C) 4
- D) 5

6

Section B (90 marks)

Question 11 (Begin and label a new booklet.) **(15 marks)**

a) Use integration by parts to find $\int \ln(x^2 + 1) dx$. [3]

b) Use the substitution $u = e^x - 1$ to find $\int \frac{e^{2x}}{(e^x - 1)^2} dx$. [3]

c) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{3 \cos x - 4 \sin x + 5} dx$ [4]

d) Convert the line $y = 3x - 5$ to the form $\underline{r} = \underline{a} + \lambda \underline{b}$. [2]

e) It is known that $5 + 6i$ is a zero of the polynomial $P(z) = 2z^3 - 19z^2 + 112z + k$, where k is real.

i) Find the other two zeroes of $P(z)$. [2]

ii) Find the value of k . [1]

(End of Question 11.)

Question 12 (Begin and label a new booklet.) **(15 marks)**

- a) Prove that $\sqrt[3]{2}$ is irrational. [3]
- b) Given that x is a real number and $(x + i)^4$ is purely imaginary, find all possible values of x in exact form. [3]
- c) Find all the solutions of $z^5 = 4 + 4i$, giving your answers in modulus-argument form. [3]
- d) i) On an Argand diagram sketch the locus of point P representing the complex number z such that $\left| z - (\sqrt{3} + i) \right| = 1$. [1]
- ii) Find the maximum possible values of $|z|$ and $\text{Arg}(z)$. [2]
- e) Prove that for all $x, y \in \mathbb{R}$,
- $$\frac{(x-1)^2 + (y+2)^2}{2} \geq x + y . \quad \text{[3]}$$

(End of Question 12.)

Question 13 (Begin and label a new booklet.) **(14 marks)**

- a) A particle is undergoing simple harmonic motion centred about the origin.

Its equation of motion is given by $\ddot{x} = -16x$.

- i) Show that $v^2 = 16(a^2 - x^2)$, where a is the amplitude. [2]

The particle's position at time t seconds is given by: $x = \sqrt{3} \sin 4t - \cos 4t$.

- ii) Express x in the form $R \sin(4t - \alpha)$, where α is in radians. [2]

- iii) When does the particle first reach its maximum speed after $t = 0$? [2]

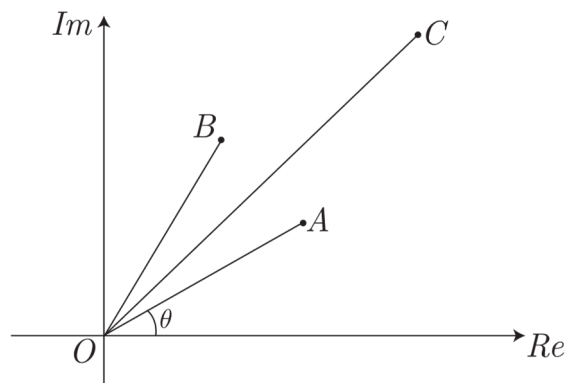
- b) i) If $z = \cos \theta + i \sin \theta$, show that $z - \frac{1}{z} = 2i \sin \theta$ and $z^n - \frac{1}{z^n} = 2i \sin n\theta$. [2]

- ii) Hence show that $\sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$. [2]

(Question 13 continues on the next page...)

- c) Let $z = e^{i\theta}$ where $0 < \theta < \frac{\pi}{2}$.

On the Argand diagram the point A represents z , the point B represents z^2 and the point C represents $z + z^2$.



Copy the diagram into your writing booklet.

- i) What type of quadrilateral best describes $OACB$? [1]
- ii) Find $\arg(z + z^2)$ in terms of θ . [1]
- iii) Find $|z + z^2|$ in terms of θ . [2]

(End of Question 13.)

Question 14 (Begin and label a new booklet.) **(14 marks)**

a) i) Prove that $x + y \geq 2\sqrt{xy}$ for positive real numbers x, y . [1]

ii) Hence for positive real numbers a, b, c , and d , prove that: [2]

$$a^2 + 4b^2 + 4c^2 + d^2 \geq 8\sqrt{abcd}$$

b) Consider the line with equation $\vec{r} = \begin{bmatrix} -5 \\ 1 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$.

Let A be the point corresponding to $\lambda = 1$ on the line.

Let $P(6, 0, 2)$ be a point not on the line.

i) Find the projection of $\overrightarrow{AP}$ onto the line. [2]

ii) Hence find the coordinates of point P' , the reflection of P about the line. [2]

c) i) Use De Moivre's theorem to show that $\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$ and $\sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta$. [2]

ii) Deduce that $\tan 3\theta = \frac{\tan^3 \theta - 3\tan \theta}{3\tan^2 \theta - 1}$. [1]

iii) By letting $x = \tan \theta$, find the solutions of $x^3 - 3x^2 - 3x + 1 = 0$, expressing them in exact trigonometric form. [3]

iv) Hence show that $\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4$. [1]

(End of Question 14.)

Question 15 (Begin and label a new booklet.) **(16 marks)**

a) Evaluate $\int_{-1}^2 \sqrt{8-2x-x^2} \, dx$ using a suitable substitution. [3]

b) Prove the following statement for all integers $n \geq 3$.

“If $3^n - 4$ is a prime, then n is odd.” [3]

c) A particle of mass m kilograms is dropped from rest in a medium where the resistance to motion has magnitude $\frac{1}{10}mv^2$ Newtons when the velocity of the particle is v m/s. After t seconds, the particle has fallen x metres. Take $g = 10 \text{ m/s}^2$.

i) With a diagram, show that the acceleration is $\ddot{x} = \frac{1}{10}(100 - v^2)$. [1]

The particle hits the ground $\log_e(\sqrt{5})$ seconds after it is dropped.

ii) Find its distance fallen in simplest exact form. [5]

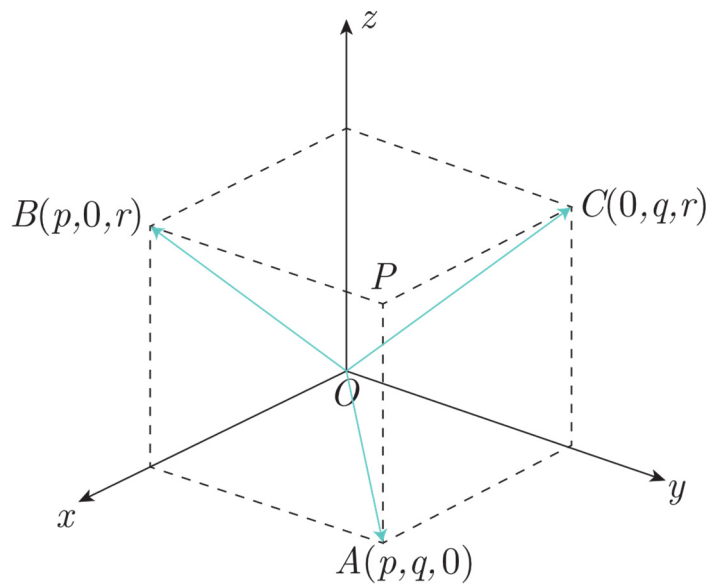
d) Let $I_n = \int_0^1 \frac{x^n}{\sqrt{1+x^2}} \, dx$ for integers $n \geq 0$.

Show that $I_n = \frac{\sqrt{2} - (n-1)I_{n-2}}{n}$ for all integers $n \geq 2$ [4]

(End of Question 15.)

Question 16 (Begin and label a new booklet.) **(16 marks)**

- a) The diagram below shows a rectangular prism in the first octant created from $P(p, q, r)$ with all edges parallel to the axes and with OA , OB and OC as face diagonals.



It is known that $\angle AOB = \angle AOC = \angle BOC = \frac{\pi}{3}$.

Prove that the rectangular prism is a cube.

[4]

(Question 16 continues on the next page...)

b) i) Show that for $k \geq 0$, $2k + 3 > 2\sqrt{(k+1)(k+2)}$. [2]

ii) Hence show that for $n \geq 1$,

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - 1). \quad [4]$$

iii) Is the statement $\sum_{k=1}^N \frac{1}{\sqrt{k}} < 10^{10}$ true for all positive integers N ? Justify your answer. [1]

c) Let z_1 and z_2 be two non-zero complex numbers.

It is known that $z_1 + z_2 + 1 = 0$.

i) Show that $|z_1| = |z_2| = 1 \Rightarrow \frac{1}{z_1} + \frac{1}{z_2} + 1 = 0$. [2]

ii) Also show that $\frac{1}{z_1} + \frac{1}{z_2} + 1 = 0 \Rightarrow |z_1| = |z_2| = 1$. [2]

iii) Hence or otherwise show that $|z_1| = |z_2| = 1 \Rightarrow z_1^2 + z_2^2 + 1 = 0$. [1]

(End of Question 16.)

End of exam.

Ascham 2025 Ext 2 Trial Solutions

MC

Q1. $\sqrt{\cos^2 \theta + \sin^2 \theta + \tan^2 \theta}$
 $= \sqrt{1 + \tan^2 \theta}$
 $= \sqrt{\sec^2 \theta}$
 $= \sec \theta$ as θ in 1st quad.

D

Q2. $\neg(p \rightarrow q) = p \wedge \neg q$

Snow is falling and I will be...

C

Q3. $\ddot{x} = -4x$

$n=2$ centre: $x=0$

$t=0: x=3$ A or B

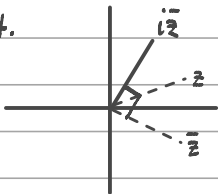
$t=0: \dot{x} = -6\sqrt{3}$

A: $\dot{x} = -6\sin 2t - 6\sqrt{3}\cos 2t$

at $t=0: \dot{x} = -6\sqrt{3}$ ✓

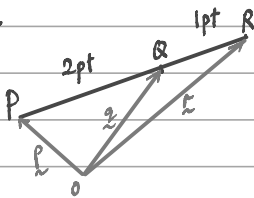
A

Q4.



A

Q5.



$$\begin{aligned}\vec{OR} &= \vec{OP} + \vec{PR} \\ &= p + \frac{3}{2}\vec{PQ} \\ &= p + \frac{3}{2}(q-p) \\ &= p + \frac{3}{2}q - \frac{3}{2}p \\ &= -\frac{1}{2}p + \frac{3}{2}q\end{aligned}$$

C

Alternatively: $\vec{QR} = \frac{1}{2}\vec{PQ}$
 $r - q = \frac{1}{2}(q - p)$
 $r = \frac{3}{2}q - \frac{1}{2}p$

Q6. Checking each converse:

$x+y$ even: $4+6$ X

x^2+y^2 even: 2^2+4^2 X

xy odd: can't think of counterexample...

x^y odd: 3^4 is odd but y even X

C

Q7. A and C out

due to $x=1$ always.

B: try $\lambda = -1$:

$$\tilde{L} = \begin{pmatrix} 1 & -1 \\ 1 & -0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \checkmark$$

D: try $\lambda = -1$:

$$\tilde{L} = \begin{pmatrix} 1 & -1 \\ 0 & -1 \\ 1 & -0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \times \quad \boxed{B}$$

Q8. A: $\frac{1}{1+x} < \frac{1}{x}$ in $[1, 2]$ $\checkmark$

B: $\sin x < 1$ in $[\frac{\pi}{6}, \frac{\pi}{4}]$ $\checkmark$

C: e^{-x^2} decreases

from $[0, 1]$ to $[1, 2]$ $\checkmark$

D: as $\tan x \leq 1$ in $[0, \frac{\pi}{4}]$

$\tan^2 x > \tan^3 x$ in $[0, \frac{\pi}{4}]$ FALSE!

$\boxed{D}$

Q9. All radius 1.

As t goes through $(0, 2\pi)$

A) X not linear increase in z -value.

C) $X \ln(t)$ would be large negative.

D) X z -value does not go up and down.

as $t \rightarrow \infty$, $z \rightarrow 0$ $\therefore$ B

$\boxed{B}$

Q10. $w^5 = e^{i2\pi} = 1$

and $1 + w + w^2 + w^3 + w^4 = 0$, ($w \neq 1$)

$$= (1 - w^5 - w + w^3)(1 - w^3)(1 - w^4)$$

$$= (1 - w^5 - w^1 + w^3 - w + w^4 + w^3 - w^4)(1 - w^4)$$

$$= (2 - w^5 - 2w + w^4)(1 - w^4)$$

$$= 2 - 2w^4 - w^5 + w^6 - 2w + 2w^5 + w^4 - w^8$$

$$= 4 - w - w^2 - w^3 - w^4$$

$$= 4 - (-1)$$

$$= 5$$

$\boxed{D}$

Alternatively: $w^4 = e^{i\frac{8\pi}{5}} = e^{i(-\frac{2\pi}{5})} = \bar{w}$

$$w^3 = e^{i\frac{6\pi}{5}} = e^{i(-\frac{4\pi}{5})} = \bar{w}^2$$

$$(1 - w)(1 - \bar{w})(1 - w^3)(1 - \bar{w}^2)$$

$$= (1 - (w + \bar{w}) + w\bar{w})(1 - (w^3 + \bar{w}^2) + w^3\bar{w}^2)$$

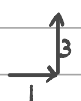
$$= (1 - 2\cos\frac{2\pi}{5} + 1)(1 - 2\cos\frac{4\pi}{5} + 1)$$

$$= 5 \text{ (with calculator...)}$$

Q11 a) $\int \ln(x^2+1) dx$ $u = \ln(x^2+1)$ $v = x$
 $= x \ln(x^2+1) - \int x \cdot \frac{2x}{x^2+1} dx$ $u' = \frac{2x}{x^2+1}$ $v' = 1$
 $= x \ln(x^2+1) - 2 \int \frac{x^2+1}{x^2+1} - \frac{1}{x^2+1} dx$
 $= x \ln(x^2+1) - 2[x - \tan^{-1}x] + C$
 $= x \ln(x^2+1) - 2x + 2\tan^{-1}x + C$

b) $\int \frac{e^{2x}}{(e^x-1)^2} dx$ $u = e^x - 1$
 $e^x = u + 1$
 $x = \ln(u+1)$
 $dx = \frac{1}{u+1} du$
 $= \int \frac{(u+1)^2}{u^2} \cdot \frac{1}{u+1} du$
 $= \int \frac{u+1}{u^2} du$
 $= \int \frac{1}{u} + \frac{1}{u^2} du$
 $= \ln|u| - \frac{1}{u} + C$
 $= \ln|e^x-1| - \frac{1}{e^x-1} + C$

c) $\int_0^{\frac{\pi}{2}} \frac{1}{3\cos x - 4\sin x + 5} dx$ $t = \tan \frac{x}{2}$
 $x=0, t=0$
 $x=\frac{\pi}{2}, t=1$
 $= \int_0^1 \frac{1}{3 \frac{1-t^2}{1+t^2} - \frac{4(2t)}{1+t^2} + 5} \cdot \frac{2}{1+t^2} dt$
 $= \int_0^1 \frac{2}{3-3t^2-8t+5+5t^2} dt$ $x = 2\tan^{-1}t$
 $dx = \frac{2}{1+t^2} dt$
 $= \int_0^1 \frac{2}{8+2t^2-8t} dt$
 $= \int_0^1 \frac{1}{t^2-4t+4} dt$
 $= \int_0^1 (t-2)^{-2} dt$
 $= \left[-(t-2)^{-1} \right]_0^1$
 $= -\frac{1}{-1} + \frac{1}{-2}$
 $= \frac{1}{2}$

d) $y = 3x - 5$
 $m = \frac{3}{1}$  $b = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$
 $y\text{-int: } (0, -5)$ $\hat{a} = \begin{pmatrix} 0 \\ -5 \end{pmatrix}$ or equiv.
 $\therefore \hat{r} = \begin{pmatrix} 0 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

e) i) $5-6i$ also a zero as coeff are real by conjugate root theorem.
 $\text{I roots: } 5+6i + 5-6i + \alpha = \frac{19}{2}$
 $10 + \alpha = \frac{19}{2}$
 $\alpha = \frac{-1}{2}$

ii) Prod roots: $(5+6i)(5-6i) + \frac{-1}{2} = \frac{-k}{2}$
 $25 + 36 = k$
 $k = 61$

Q12

a) Assume $\sqrt[3]{2} = \frac{p}{q}$, $p, q \in \mathbb{Z}$ and $\gcd(p, q) = 1$.

$$\sqrt[3]{2} q = p$$

$$2q^3 = p^3$$

$\therefore p^3$ is divisible by 2

$\therefore p$ is divisible by 2 by considering the contrapositive

$\therefore$ Let $p = 2m$, $m \in \mathbb{Z}$

$$2q^3 = (2m)^3$$

$$2q^3 = 8m^3$$

$$q^3 = 4m^3$$

$\therefore q^3$ is divisible by 2

$\therefore q$ is divisible by 2 by contrapositive

But $\gcd(p, q)$ is 1 from assumption

$\therefore \sqrt[3]{2}$ is irrational by contradiction.

$$b) (x+i)^4 = {}^4C_0 x^4 + {}^4C_1 x^3 i + {}^4C_2 x^2 i^2 + {}^4C_3 x i^3 + {}^4C_4 i^4$$

$$\operatorname{Re}(x+i)^4 = 0 :$$

$$x^4 - 6x^2 + 1 = 0$$

$$x^2 = \frac{6 \pm \sqrt{36-4}}{2}$$

$$= \frac{6 \pm 4\sqrt{2}}{2}$$

$$= 3 \pm 2\sqrt{2}$$

$$\therefore x = \pm \sqrt{3 \pm 2\sqrt{2}}$$

$$c) 4+4i = 4\sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

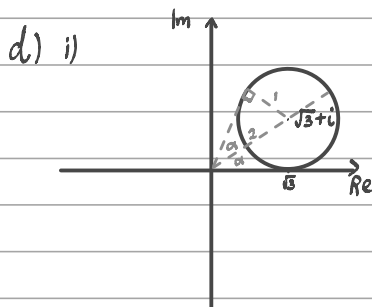
$$z = r \operatorname{cis} \theta$$

$$z^5 = r^5 \operatorname{cis} 5\theta$$

$$r^5 = 4\sqrt{2} \quad 5\theta = \frac{\pi}{4} + 2k\pi, \quad k=0,1,2,3,4$$

$$r = \sqrt{2}$$

$$\therefore z = \sqrt{2} \operatorname{cis} \left(\frac{\pi}{20} + \frac{2k\pi}{5} \right), \quad k=0,1,2,3,4$$



$$ii) \operatorname{Max} |z| = |\sqrt{3}+i| + 1$$

$$= 2+1$$

$$= 3$$

$$\operatorname{Max} \operatorname{Arg}(z) = 2\alpha$$

$$= 2 \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{3}$$

$$e) \text{ RTP: } \frac{x^2-2x+1+y^2+4y+4}{2} - x - y \geq 0$$

$$\text{LHS} = \frac{x^2-4x+4+y^2+2y+1}{2}$$

$$= \frac{1}{2}(x-2)^2 + \frac{1}{2}(y+1)^2$$

$$\geq 0 \text{ as perfect squares } \geq 0$$

Q13

a) i) $\dot{x} = -16x$

$$v \frac{dv}{dx} = -16x$$

$$\int v dv = \int -16x dx$$

$$\frac{1}{2}v^2 = -8x^2 + c$$

$$v^2 = -16x^2 + c_1$$

$$v=0, x=a$$

$$0 = -16a^2 + c_1$$

$$16a = c_1$$

$$\therefore v^2 = -16x^2 + 16a^2$$

$$v^2 = 16(a^2 - x^2)$$

ii) $x = \sqrt{3} \sin 4t - \cos 4t$

$$x = R \sin(4t - \alpha)$$

$$= R (\sin 4t \cos \alpha - \cos 4t \sin \alpha)$$

$$\therefore \sqrt{3} = R \cos \alpha \quad 1 = R \sin \alpha$$

$$R^2 = \sqrt{3}^2 + 1^2$$

$$= 4$$

$$R = 2$$

$$\therefore x = 2 \sin(4t - \frac{\pi}{6})$$

$$\therefore \frac{\sqrt{3}}{2} = \cos \alpha$$

$$\text{rel } \angle = \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{6}$$

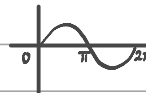
iii) Max speed at centre $x=0$

$$0 = 2 \sin(4t - \frac{\pi}{6})$$

$$4t - \frac{\pi}{6} = 0, \pi, 2\pi \dots$$

$$4t = \frac{\pi}{6}$$

$$t = \frac{\pi}{24} \text{ sec}$$



b) i) $z = \cos \theta + i \sin \theta \quad |z|=1 \Rightarrow \frac{1}{z} = \bar{z}$

$$z - \frac{1}{z} = \cos \theta + i \sin \theta - (\cos \theta - i \sin \theta)$$

$$= 2i \sin \theta$$

$$z^n = \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = \bar{z}^n = \cos n\theta - i \sin n\theta$$

$$\therefore z^n - \frac{1}{z^n} = 2i \sin n\theta$$

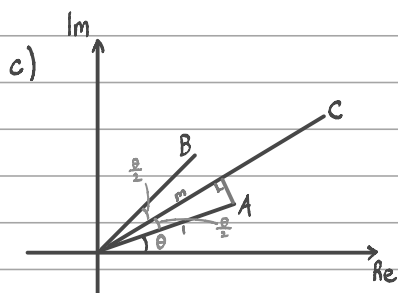
ii) $\sin 5\theta = \left(\frac{1}{2i} \left(z - \frac{1}{z}\right)\right)^5$

$$= \frac{1}{32i} \left(z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}\right)$$

$$= \frac{1}{32i} \left(z^5 - \frac{1}{z^5} - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)\right)$$

$$= \frac{1}{32i} \left(2i \sin 5\theta - 5 \times 2i \sin 3\theta + 10 \times 2i \sin \theta\right)$$

$$= \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$$



i) Rhombus since $|z|=1$

so $|z^2|=1$ and parallelogram has equal sides.

ii) $\arg(z + z^2) = \theta + \frac{\theta}{2} = \frac{3\theta}{2}$

iii) On diagram, $\frac{m}{1} = \cos \frac{\theta}{2}$

$$\therefore |z + z^2| = |\vec{OC}| = 2 \cos \frac{\theta}{2}$$

Q14

a) i) $(a-b)^2 \geq 0$
 $a^2 - 2ab + b^2 \geq 0$
 $a^2 + b^2 \geq 2ab$
 $a \rightarrow \sqrt{x}, b \rightarrow \sqrt{y}$
 $\therefore x + y \geq 2\sqrt{xy}$

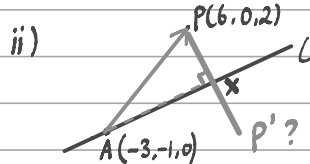
ii) using AM $\geq$ GM.

$$\begin{aligned} a^2 + 4b^2 &\geq 2\sqrt{4a^2b^2} = 4ab \\ 4c^2 + d^2 &\geq 2\sqrt{4c^2d^2} = 4cd \\ \therefore a^2 + 4b^2 + 4c^2 + d^2 &\geq 4ab + 4cd \\ &\geq 2\sqrt{16abcd} \\ &= 8\sqrt{abcd} \end{aligned}$$

b) i) $A: \begin{pmatrix} -5+2 \\ 1-2 \\ -1+1 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 0 \end{pmatrix}$

$$\begin{aligned} \vec{AP} &= \begin{pmatrix} 6 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} -3 \\ -1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 9 \\ 1 \\ 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{proj}_{\vec{AP}} \vec{AP} &= \frac{\begin{pmatrix} 9 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}}{\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \\ &= \frac{18 - 2 + 2}{2^2 + 2^2 + 1^2} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \\ &= 2 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} \end{aligned}$$



Let $\vec{AX}$ be $\text{proj}_{\vec{AP}} \vec{AP}$

$$\begin{aligned} \therefore \vec{PX} &= \vec{AX} - \vec{AP} \\ &= \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} - \begin{pmatrix} 9 \\ 1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} -5 \\ -5 \\ 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \therefore \vec{PP'} &= 2\vec{PX} \\ &= \begin{pmatrix} -10 \\ -10 \\ 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \therefore \vec{OP'} &= \vec{OP} + \vec{PP'} \\ &= \begin{pmatrix} 6 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} -10 \\ -10 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} -4 \\ -10 \\ 2 \end{pmatrix} \therefore P'(-4, -10, 2) \end{aligned}$$

c) i) $(\cos \theta + i \sin \theta)^3 = c^3 + 3c^2si + 3cs^2i^2 + s^3i^3$
 where $c = \cos \theta$ and $s = \sin \theta$

Using De Moivre's: LHS = $\cos 3\theta + i \sin 3\theta$

Equating Re: $\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$

Equating Im: $\sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta$

$$\begin{aligned} \text{ii) } \tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\cos \theta \sin^2 \theta} \div \cos^3 \theta \\ &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \\ &= \frac{\tan^3 \theta - 3 \tan \theta}{3 \tan^2 \theta - 1} \end{aligned}$$

iii) $x = \tan \theta : x^3 - 3x^2 - 3x + 1 = 0$

$$\tan^3 \theta - 3 \tan^2 \theta - 3 \tan \theta + 1 = 0$$

$$\tan^3 \theta - 3 \tan \theta = 3 \tan^2 \theta - 1$$

$$\frac{\tan^3 \theta - 3 \tan \theta}{3 \tan^2 \theta - 1} = 1$$

$$\tan 3\theta = 1 \quad \text{rel } \angle = \frac{\pi}{4} \quad \text{✗}$$

$$3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

$$\therefore x = \tan \frac{\pi}{12}, \tan \frac{5\pi}{12}, \tan \frac{3\pi}{4} = -1$$

iv) $\sum \text{roots} : \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} - 1 = \frac{-3}{1}$

$$\therefore \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4$$

Q15

a) $\int_{-1}^2 \sqrt{8-2x-x^2} dx$

$$= \int_{-1}^2 \sqrt{8-(x^2+2x+1)-1} dx$$

$$= \int_{-1}^2 \sqrt{9-(x+1)^2} dx$$

$$x+1 = 3 \sin \theta$$

$$x=2 : 3 = 3 \sin \theta$$

$$x = 3 \sin \theta - 1$$

$$\theta = \frac{\pi}{2}$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{9-9\sin^2 \theta} \cdot 3 \cos \theta d\theta$$

$$dx = 3 \cos \theta d\theta$$

$$x=-1 : 0 = 3 \sin \theta$$

$$\theta = 0$$

$$= \int_0^{\frac{\pi}{2}} 9 \cos^2 \theta d\theta \text{ as } |\cos \theta| = \cos \theta \text{ for } \theta \in [0, \frac{\pi}{2}]$$

$$= 9 \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos 2\theta d\theta$$

$$= \frac{9}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{9}{2} \left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \frac{9}{2} (0+0)$$

$$= \frac{9\pi}{4}$$

b) Contrapositive: if n is even, then $3^n - 4$ is not prime.

Let $n = 2m$, $m \in \mathbb{Z}^+$

$$3^{2m} - 4 = (3^m)^2$$

$$= (3^m - 2)(3^m + 2)$$

Since $m \geq 2$, $3^m - 2$ and $3^m + 2$ are integers ≥ 2

$\therefore 3^n - 4$ is not prime as it has more factors other than 1 and itself.

c) i)

$$\begin{array}{c} + \downarrow \\ \downarrow \frac{1}{10}mv^2 \\ \downarrow mg \end{array}$$

$$\therefore F_{\text{net}} = mg - \frac{1}{10}mv^2$$

$$ma = 10m - \frac{1}{10}mv^2$$

$$a = 10 - \frac{1}{10}v^2$$

$$= \frac{1}{10}(100 - v^2)$$

ii) $v \frac{dv}{dx} = \frac{1}{10}(100 - v^2)$

$$\frac{1}{2} \int \frac{-2v}{100 - v^2} dv = \frac{1}{10} \int dx$$

$$\frac{1}{2} \ln |100 - v^2| = \frac{x}{10} + c$$

$$x = 0, v = 0 :$$

$$\frac{1}{2} \ln 100 = c$$

$$\frac{1}{2} \ln |100 - v^2| = \frac{x}{10} - \frac{1}{2} \ln 100$$

$$\frac{1}{2} \ln \left| \frac{100 - v^2}{100} \right| = \frac{x}{10}$$

$$-5 \ln \left| \frac{100 - v^2}{100} \right| = x$$

needs $v \dots$ see right.

$$\therefore \text{at } v = \frac{20}{3}$$

$$x = -5 \ln \left| \frac{100 - (\frac{20}{3})^2}{100} \right|$$

$$= -5 \ln \left(\frac{5}{9} \right)$$

$$= 5 \ln \frac{9}{5} \text{ metres}$$

$$\frac{dv}{dt} = \frac{1}{10}(100 - v^2)$$

$$\int \frac{1}{(10-v)(10+v)} dv = \int \frac{1}{10} dt$$

$$\frac{1}{20} \int \left(\frac{-1}{10-v} + \frac{1}{10+v} \right) dv = \int \frac{1}{10} dt$$

$$\frac{1}{20} (-\ln |10-v| + \ln |10+v|) = \frac{t}{10} + c$$

$$\frac{1}{20} \ln \left| \frac{10+v}{10-v} \right| = \frac{t}{10} + c$$

$$\frac{1}{20} \ln 1 = 0 + c \therefore c = 0$$

$$\therefore \frac{1}{20} \ln \left| \frac{10+v}{10-v} \right| = \frac{t}{10}$$

$$\text{at } t = \ln(\sqrt{5}) :$$

$$\frac{1}{20} \ln \left| \frac{10+v}{10-v} \right| = \ln \sqrt{5} = \frac{1}{2} \ln 5$$

$$\left| \frac{10+v}{10-v} \right| = 5$$

$$\text{as } \ddot{x} = 0 \Rightarrow 100 - v^2 = 0$$

$v = 10$ is terminal vel.

$$0 \leq v < 10 \Rightarrow \left| \frac{10+v}{10-v} \right| = \frac{10+v}{10-v} = 5$$

$$10+v = 50-5v$$

$$6v = 40$$

$$v = \frac{20}{3}$$



$$15d) I_n = \int_0^1 x^{n-1} x (1+x^2)^{-\frac{1}{2}} dx \quad u = x^{n-1} \quad v = \sqrt{1+x^2}$$

$$= \left[x^{n-1} \sqrt{1+x^2} \right]_0^1 - \int_0^1 \sqrt{1+x^2} (n-1) x^{n-2} dx \quad v' = x(1+x^2)^{-\frac{1}{2}}$$

$$= \sqrt{2} - (n-1) \int_0^1 \frac{x^{n-2} (1+x^2)}{\sqrt{1+x^2}} dx \quad v = \frac{1}{2} \int 2x(1+x^2)^{-\frac{1}{2}} dx$$

$$= \frac{1}{2} \times 2 (1+x^2)^{\frac{1}{2}}$$

$$I_n = \sqrt{2} - (n-1) [I_{n-2} + I_n]$$

$$I_n = \sqrt{2} - (n-1)I_n - (n-1)I_{n-2}$$

$$I_n(1+n-1) = \sqrt{2} - (n-1)I_{n-2}$$

$$I = \frac{\sqrt{2} - (n-1)I_{n-2}}{n} \text{ as required.}$$

Q16 a) $\angle AOB = \frac{\pi}{3}$

$$\Rightarrow \cos \frac{\pi}{3} = \frac{\vec{OA} \cdot \vec{OB}}{|\vec{OA}| |\vec{OB}|}$$

$$= \frac{\begin{pmatrix} p \\ q \\ r \end{pmatrix} \cdot \begin{pmatrix} p \\ q \\ r \end{pmatrix}}{\sqrt{p^2+q^2} \sqrt{p^2+q^2}}$$

$$\frac{1}{2} = \frac{p^2}{\sqrt{p^2+q^2} \sqrt{p^2+q^2}} \dots ①$$

Similarly: $\frac{1}{2} = \frac{q^2}{\sqrt{q^2+p^2} \sqrt{q^2+r^2}} \dots ②$

and $\frac{1}{2} = \frac{r^2}{\sqrt{r^2+p^2} \sqrt{r^2+q^2}} \dots ③$

RTP (hopefully) $p=q=r$

$$① = ②: \frac{p^2}{\sqrt{p^2+q^2} \sqrt{p^2+r^2}} = \frac{q^2}{\sqrt{q^2+p^2} \sqrt{q^2+r^2}}$$

$$p^2 \sqrt{q^2+r^2} = q^2 \sqrt{p^2+r^2}$$

$$p^4 (q^2+r^2) = q^4 (p^2+r^2)$$

$$p^4 q^2 - q^4 p^2 + p^4 r^2 - q^4 r^2 = 0$$

$$p^2 q^2 (p^2 - q^2) + r^2 (p^2 - q^2) (p^2 + q^2) = 0$$

$$(p^2 - q^2) [p^2 q^2 + r^2 (p^2 + q^2)] = 0$$

since $[...] > 0$ due to lengths,

$$p^2 = q^2 \Rightarrow p = q.$$

Similarly for $p=r$ and $q=r$.

$\therefore$ the prism is a cube.

16 b) i) RTP: $2k+3 > 2\sqrt{(k+1)(k+2)}$

(Backtrack)

$$4k^2 + 12k + 9 \geq 4(k+1)(k+2)$$

$$\text{LHS}^2 = (2k+3)^2$$

$$\text{RHS}^2 = 4(k+1)(k+2)$$

$$= 4(k^2 + 3k + 2)$$

$$= 4k^2 + 12k + 9$$

$$= 4(k^2 + 3k + 2)$$

$$= 4k^2 + 12k + 8$$

$$= 4k^2 + 12k + 8$$

$$\therefore \text{LHS}^2 \geq \text{RHS}^2$$

$$\therefore \text{LHS} \geq \text{RHS} \text{ as } \text{LHS}, \text{RHS} > 0 \text{ for } k \geq 0$$

ii) Proof by M.I.:

Test P(1): $\text{LHS} = \frac{1}{\sqrt{1}} = 1$

$$\text{RHS} = 2(\sqrt{2} - 1)$$

$$\approx 0.8$$

$$\therefore \text{LHS} > \text{RHS} \text{ True for } n=1$$

Assume P(k): $1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} > 2(\sqrt{k+1} - 1)$

RTP P(k+1): $1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} > 2(\sqrt{k+1+1} - 1) = 2(\sqrt{k+2} - 1)$

By assumption:

$$1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} > 2(\sqrt{k+1} - 1) + \frac{1}{\sqrt{k+1}}$$

$$= 2\sqrt{k+1} + \frac{1}{\sqrt{k+1}} - 2$$

$$= \frac{2(k+1)+1}{\sqrt{k+1}} - 2$$

$$= \frac{2k+3}{\sqrt{k+1}} - 2$$

$$> \frac{2\sqrt{(k+1)(k+2)}}{\sqrt{k+1}} - 2 \quad \text{from i)}$$

$$= 2\sqrt{k+2} - 2$$

$$= 2(\sqrt{k+2} - 1)$$

$\therefore$ Statement true by M.I.

iii) From ii) we know $\sum_{k=1}^N \frac{1}{\sqrt{k}} > 2(\sqrt{N+1} - 1)$

So statement is false if we choose

$$2(\sqrt{N+1} - 1) > 10^{10}$$

$$\sqrt{N+1} > \frac{10^{10}}{2} + 1$$

$$N > \left(\frac{10^{10}}{2} + 1\right)^2 - 1$$

as a counterexample.

$$16c) \text{ i) RTP: } |z_1| = |z_2| = 1 \Rightarrow \frac{1}{z_1} + \frac{1}{z_2} + 1 = 0$$

$$|z|^2 = z\bar{z} = 1 \Rightarrow \frac{1}{z} = \bar{z} \text{ and } \frac{1}{z_2} = \bar{z}_2$$

$$\begin{aligned} \therefore \text{LHS} &= \frac{1}{z_1} + \frac{1}{z_2} + 1 \\ &= \bar{z}_1 + \bar{z}_2 + 1 \text{ using} \\ &= \overline{z_1 + z_2 + 1} \\ &= 0 \text{ using given.} \end{aligned}$$

$$\text{ii) RTP: } \frac{1}{z_1} + \frac{1}{z_2} + 1 = 0 \Rightarrow |z_1| = |z_2| = 1$$

$$\frac{z_2 + z_1}{z_1 z_2} = -1$$

$$z_2 + z_1 = -z_1 z_2$$

$$\therefore -1 = -z_1(-1-z_1) \text{ using } z_1 + z_2 + 1 = 0$$

$$-1 = +z_1 + z_1^2$$

$$0 = z_1^2 + z_1 + 1$$

$\therefore z_1$ is a non-real cube root of unity.

$$\text{as } 0 = \frac{z_1^3 - 1}{z_1 - 1}$$

$\therefore |z_1| = 1$ and similarly for $|z_2| = 1$

$$\text{iii) RTP } |z_1| = |z_2| = 1 \Rightarrow z_1^2 + z_2^2 + 1 = 0$$

$$\text{from i) } |z_1| = |z_2| = 1 \Leftrightarrow \frac{1}{z_1} + \frac{1}{z_2} + 1 = 0$$

$$\begin{aligned} \text{LHS} &= z_1^2 + z_2^2 + 1 \\ &= z_1^2 + (-1-z_1)^2 + 1 \\ &= z_1^2 + 1 + 2z_1 + z_1^2 + 1 \\ &= 2z_1^2 + 2z_1 + 2 \\ &= 2(z_1^2 + z_1 + 1) \end{aligned}$$

$$\text{But } \frac{1}{z_1} + \frac{1}{z_2} + 1 = 0 \Rightarrow z_1^2 + z_1 + 1 = 0 \text{ shown in ii)}$$

$$\therefore \text{LHS} = 2 \times 0$$

$$= 0 \text{ as required.}$$